

Materials and Methods

LD decay, LD-based complexity reduction, and analyses of ancestry sorting, genomic architecture, and climate association

Poikela et al.

Draft prepared from the current /doc pipeline documentation

This is a self-contained working draft for integration into the full Supplementary Materials. Blue boxes identify editorial or computational items that remain open and are not intended as final manuscript text.

S1 Scope and analytical principles

These methods describe LD decay estimation, LD-based complexity reduction, quantification of the extent and direction of ancestry sorting, analysis of its genomic architecture, and tests of whether ancestry-diagnostic or directionally sorted loci occur preferentially in regions whose allele frequencies covary with climate.

Three principles guided the analyses. First, markers in LD carry partly redundant information. We therefore used clusters of strongly correlated markers as LD-reduced analytical units wherever possible, while retaining marker-level analyses as diagnostics of spatial pseudoreplication. We do not assume that the resulting units are strictly statistically independent. Second, a variable evaluated as a predictor was not also used to select the analysed data. In particular, loci were gated by parental allele frequency rather than by the diagnostic index (DI), leaving the full range of DI available for analysis. Third, cluster-level criteria were expressed as rates rather than fixed counts when a count would make eligibility depend non-linearly on cluster size.

The twenty hybrid populations provide repeated observations of ancestry sorting across geographically separated populations. However, their evolutionary histories need not be fully independent. We therefore refer to them as replicate or sampled hybrid populations without assuming independent origins or complete demographic independence. Analyses that use variation among populations either model their covariance explicitly or are interpreted as descriptive associations.

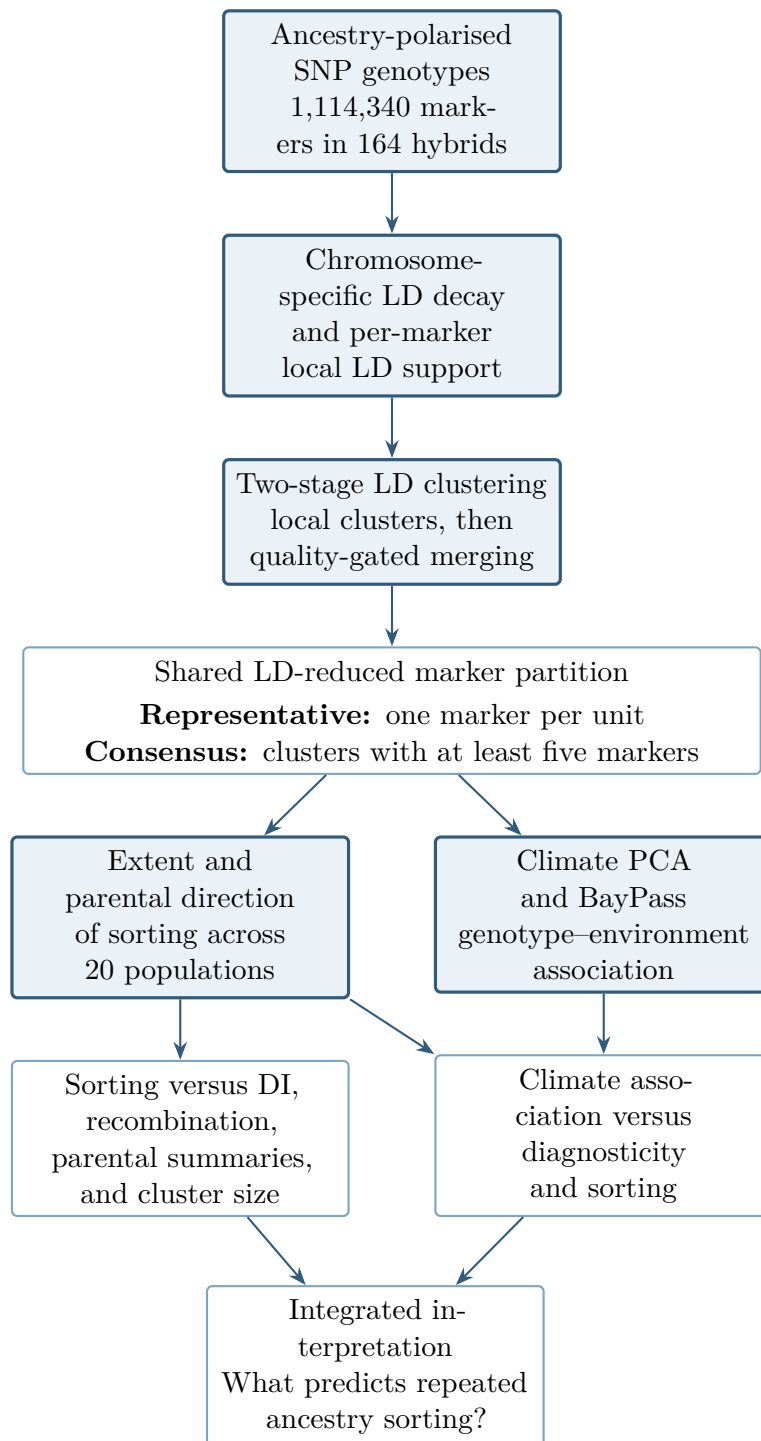


Figure 1: **Analytical workflow.** LD decay and clustering transform the marker data into a shared set of LD-reduced units represented either by a single marker or, for larger clusters, by a consensus genotype. The same units then connect the analyses of ancestry sorting, genomic architecture, and climate association.

S2 Samples, genotypes, and ancestry polarisation

S2.1 Samples

The analysis included twenty *F. aquilonia* $\times$ *F. polycтена* hybrid populations sampled from nineteen locations in Finland and one location in Switzerland, together with allopatric reference samples of the parental species. The current analysis objects contain 164 diploid hybrid individuals. The analyses described here concern this diploid hybrid data set.

Open item. The general sample description currently reports 165 sequenced hybrid individuals, whereas the analysis objects contain 164. Confirm which individual was excluded, most likely because of excessive missing data, and report the identifier and exclusion criterion in the final text. Until then, 164 is used for all analysis-specific counts.

S2.2 Marker processing and ancestry orientation

Genotypes and marker-level ancestry information were obtained from the DIEM output for this data set. Sites were restricted to biallelic markers. DIEM provides a polarity and a diagnostic index for each marker. The diagnostic index measures the degree to which allele frequencies differentiate the parental reference samples. For markers with the opposite polarity, genotype dosages were replaced by $2 - \text{dosage}$, placing dosages on a consistent ancestry-oriented scale across the genome.

Minor allele frequency (MAF) was calculated among hybrid individuals only, and markers with hybrid $\text{MAF} \geq 0.05$ were retained. The resulting analysis set contained 1,114,340 markers. Two matrices were constructed over this same marker set. The hybrids-only matrix was used for LD estimation, clustering, and complexity reduction. A second matrix containing hybrids and parental references was used when parental allele frequencies were required, including ancestry orientation and marker-level parental summaries. Keeping the parents out of the LD calculations prevents parent–hybrid differentiation from dominating estimates intended to describe LD within the hybrid populations.

S3 LD decay and local LD support

S3.1 Chromosome-specific LD decay

LD decay was modelled separately for each chromosome as

$$r^2(d) = b + \frac{c - b}{1 + ad}, \quad (1)$$

where d is physical distance in base pairs, a is the decay rate, b is long-range background r^2 , and c is short-range r^2 . Background LD was estimated from inter-chromosomal marker pairs. Within each chromosome, the curve was fitted across twenty sliding windows with 50% overlap. Up to 5,000 marker pairs were sampled per window, stratified across twenty distance classes to represent short and long distances comparably. The 0.95 quantile of r^2 was used as the within-class summary, and parameter estimates were aggregated over the central 95% of windows to reduce the influence of outlying fits (Fig. S1).

Only pairs for which both markers had $\text{MAF} > 0.1$ contributed to the background and decay-curve fits. This restriction reduces the frequency-driven depression of r^2 at rare alleles. It was

applied only during curve fitting; the stored LD edge lists retained the complete marker set used for clustering.

Decay rate varied with chromosome length. A robust regression of a on chromosome size was therefore fitted genome-wide. For a relative LD threshold ρ , the corresponding physical distance was calculated as

$$d = \frac{\rho}{a(1 - \rho)}. \quad (2)$$

This allowed window sizes to reflect each chromosome’s fitted decay behaviour rather than imposing one physical-distance threshold across the genome.

S3.2 Per-marker local LD support

For each marker, local LD support (ld_w) was calculated as the median r^2 between that marker and neighbouring markers within the window corresponding to $\rho = 0.95$. Unlike map-based recombination rate, ld_w is measured directly from the genotype data at marker resolution and requires no interpolation.

We evaluated ld_w and the window-level decay rate a as indicators of low recombination using map-based recombination rate as a reference. Across definitions based on the lowest 5%, 10%, and 25% of recombination windows, the area under the receiver-operating-characteristic curve was 0.91–0.98 for a and 0.73–0.86 for ld_w . Accordingly, a was the stronger window-level indicator, whereas ld_w supplied the marker-resolution measure needed to identify clusters for reconsideration during complexity reduction (Figs S2 and S3).

S4 Two-stage LD-based complexity reduction

S4.1 Terminology and outputs

The procedure has two stages: (1) local clustering with a complete-linkage refinement and (2) targeted, quality-gated merging. Both stages define a single partition of markers, from which two representations are produced:

- one representative marker per cluster, forming an LD-pruned marker set;
- one expected multi-locus genotype (eMLG) consensus dosage for clusters containing at least five markers.

We use “LD cluster” or “LD-reduced unit” for a set of strongly correlated markers. These units are not haplotype blocks: markers are grouped by the partition of sampled chromosomes that they report, and clusters may therefore overlap in physical position. Nor do we describe them as strictly independent. The procedure reduces redundancy and constrains residual LD; the remaining dependence is assessed or controlled where it matters downstream.

S4.2 Stage 1: local clustering

For each chromosome, the clustering distance was derived from its fitted LD decay curve at $\rho = 0.5$. The threshold was not allowed to vary further within chromosomes, because locally elevated LD is precisely where redundant markers should be grouped more strongly.

Clustering was performed on the precomputed sparse LD graph. Markers were first assigned to connected components of the thresholded graph. Each component was then refined by complete-linkage hierarchical clustering using the dense pairwise r^2 matrix within the component. The complete-linkage step limits single-linkage chaining, in which two weakly correlated markers could otherwise be joined through an intermediate marker correlated with both.

The representative marker of a cluster was the member with the highest median r^2 to the other cluster members.

S4.3 Stage 2: targeted, quality-gated merging

The finite window used to construct the Stage 1 edge list makes the first stage conservative in extended low-recombination regions: marker pairs outside the window are not evaluated and therefore cannot satisfy complete linkage. Stage 2 reconsidered candidate clusters directly from their genotypes, without the Stage 1 window restriction.

A Stage 1 cluster was flagged if any member had $ld_w > 0.025$, or if the cluster contained at least five markers. Flagged clusters were considered for merging using a distance-restricted, average-linkage dynamic tree cut. A candidate merge was accepted only while both of the following criteria were satisfied:

1. consensus fidelity

$$\text{score}_{\text{eMLG}}(x) = \text{cor}\{\text{round}(x), x\}^2 \geq 0.80;$$

2. a pairwise correlation floor of $r^2 \geq 0.2$ between the sides being merged.

The fidelity criterion evaluates whether the resulting consensus remains close to a discrete genotype. The correlation floor prevents a small, weakly related set of markers from being absorbed into a large cluster whose existing consensus would otherwise dominate the fidelity score. These gates constrain merging but do not establish complete statistical independence among the final units. Observed residual correlations among neighbouring merged units were of order $r^2 \approx 0.3$ after correction for relatedness and were local in genetic distance.

Candidate merges were restricted to 0.5 cM using the genetic map. Genetic rather than physical distance was used because a common physical distance represents different recombination distances across chromosomes and genomic regions.

S4.4 Output and validation

The final partition contained 474,014 LD clusters. Of these, 32,840 contained at least five markers and had an eMLG consensus. Smaller clusters remain valid LD-reduced units and retain their representative marker; the five-marker threshold is therefore a threshold for storing a consensus, not for defining a cluster.

The procedure was evaluated by comparing Stage 1 and combined Stage 1–2 clusterings in extended-LD regions (Fig. S4), by examining the distribution of consensus fidelity across ld_w thresholds (Fig. S5), and by quantifying the ability of ld_w to identify low-recombination regions. These checks assess whether Stage 2 consolidates fragmentation where long-range LD is present without broadly collapsing weakly correlated markers.

S5 Extent and direction of ancestry sorting

S5.1 Near-fixation and ancestry orientation

For each marker or cluster consensus, dosage was oriented as the *F. aquilonia*-allele frequency using the sign of the parental allele-frequency difference. Units for which the parents did not differ had no defined parental direction. Within each hybrid population, a unit was classified as near-fixed toward *F. aquilonia* when the oriented allele frequency was at least $1 - \theta$, near-fixed toward *F. polychaeta* when it was at most θ , and otherwise not near-fixed, with $\theta = 0.15$. The same tolerance was used throughout because the attainable frequency values depend on the number of sampled diploid individuals in a population.

S5.2 Sorting decomposition and classification

Let n_{obs} be the number of populations with an observed and oriented frequency, and n_{aqu} and n_{pol} the numbers near-fixed toward *F. aquilonia* and *F. polychaeta*, respectively. The total proportion near-fixed was decomposed as

$$\frac{n_{\text{aqu}} + n_{\text{pol}}}{n_{\text{obs}}} = \left| \frac{n_{\text{aqu}} - n_{\text{pol}}}{n_{\text{obs}}} \right| + \frac{2 \min(n_{\text{aqu}}, n_{\text{pol}})}{n_{\text{obs}}}, \quad (3)$$

where the first term is the unidirectional component and the second the bidirectional component. With $\tau = 0.5$, a unit was classified as unidirectional when the absolute unidirectional component was at least the bidirectional component and at least τ ; bidirectional when the bidirectional component exceeded the absolute unidirectional component and was at least τ ; and unsorted otherwise. Exchanging the parental labels changes the direction but not the magnitude or class.

Nouhaud et al. [1] evaluated predictable ancestry sorting by asking whether the same ancestry component repeatedly fixed across three hybrid populations and summarising the outcome in fixed 20-kb windows. The present classification retains that biological contrast but extends it to twenty populations, estimates the outcome directly from population allele frequencies, and separates consistent-direction from opposing-direction near-fixation. Applying the statistic first to SNPs and then to LD-reduced units also avoids assigning equal analytical status to physical windows that may contain very different amounts of redundant information.

S5.3 Conditional test of directional predictability

Among populations in which a unit was near-fixed, n_{aqu} was compared with $\text{Binomial}(n_{\text{fixed}}, 0.5)$ using a two-sided exact test. Tests were calculated only for units near-fixed in at least five populations, and Benjamini–Hochberg adjustment was applied across the tested units. The classification and test were kept separate: the classification describes the extent and direction of sorting at fixed thresholds, whereas the test evaluates departure from an equal-direction expectation conditional on the observed number of near-fixed populations.

These P -values are interpreted as conditional, exploratory evidence rather than as stand-alone confirmation of selection. The population panel may retain shared demographic structure, the tests are performed after eligibility filters, and the biological interpretation of any strong candidate requires support from genomic context and independent biological evidence.

S5.4 Parental polymorphism gate

Near-fixation was interpreted as sorting only for variants segregating in the founding parental pool. Analyses were therefore restricted to units with pooled parental MAF ≥ 0.15 , calculated from pooled parental allele counts and folded to the minor allele. This gate excludes variants already close to monomorphic in the parental pool without selecting on DI, which is evaluated as a predictor in subsequent analyses.

S5.5 Cluster-level analysis and dilution check

The sorting statistic was first calculated per marker. It was then recalculated for eMLG consensus genotypes so that a cluster contributed one aggregate observation. For clusters without a stored consensus but containing a marker-level sorting signal, a consensus was constructed by the same procedure used for larger clusters. Parental consensus genotypes reused the reference marker and allele-flip decisions derived from the hybrid genotypes, preventing hybrid and parental consensus from being placed on inconsistent orientations. Cluster DI was aggregated as the maximum DI among member markers.

The cluster-level aggregation analysis is conditional: clusters entered this comparison because they contained at least one marker with a SNP-level sorting signal. Consequently, the proportion of aggregated clusters classified as unsorted does not estimate the genome-wide fraction of unsorted clusters. Instead, it asks whether an apparent marker-level sorting signal remains after correlated members are represented jointly. The resulting comparison is therefore conditional rather than an estimate of the genome-wide frequency of unsorted clusters.

When a cluster containing an individually sorted marker was unsorted after aggregation, consensus fidelity was used to distinguish lack of an aggregate signal from possible dilution in a heterogeneous consensus. Low-fidelity clusters were flagged rather than automatically interpreted as evidence against sorting.

S5.6 Models of sorting direction

Among unidirectionally sorted units, the probability of sorting toward *F. aquilonia* was modelled by logistic regression. The first model included DI and pooled parental MAF. A second model additionally included $\log_2(\text{cluster size})$ to test whether the apparent relationship between cluster size and direction persisted after accounting for DI. Associations are interpreted descriptively and are not assigned a causal mechanism from these regressions alone.

S6 Genomic architecture of sorting

S6.1 Marker-level genomic summaries

Map-based recombination rate was assigned to markers by linear interpolation and expressed in cM/Mb. From the parental allele frequencies p_{aqu} and p_{pol} , we calculated:

- mean parental expected heterozygosity, obtained by averaging the two species-specific expected heterozygosities;
- the between-parent allelic-difference summary

$$d_{xy} = p_{\text{aqu}}(1 - p_{\text{pol}}) + p_{\text{pol}}(1 - p_{\text{aqu}});$$

- relative differentiation, $F_{ST} = (H_T - H_S)/H_T$.

These are marker-level summaries conditional on the SNPs retained for analysis. They should not be interpreted as genome-wide nucleotide diversity or absolute sequence divergence because invariant sites are not included. In the final manuscript we therefore use the restricted terms “parental expected heterozygosity”, “between-parent allelic difference at analysed SNPs”, and “relative differentiation at analysed SNPs”.

S6.2 Marker-level and LD-reduced summaries

Sorting was summarised across recombination-rate deciles twice: once using one representative per LD-reduced unit and once using a random sample of individual markers. This comparison was used to diagnose spatial pseudoreplication. Differences between the two summaries indicate that a marker-level pattern is influenced by unequal redundancy across the recombination landscape.

S6.3 Association models

Relationships among DI, recombination rate, parental expected heterozygosity, between-parent allelic difference, relative differentiation, pooled parental MAF, and cluster size were first summarised using rank correlations and recombination deciles (Fig. S6).

The magnitude of sorting was modelled against recombination rate at the LD-reduced-unit level. Cluster size was not included in this model because it can be a consequence of both recombination and local marker variability; conditioning on it could induce an association between its causes.

Among unidirectionally sorted units, sorting direction was analysed by logistic regression with standardised DI, log recombination rate, pooled parental MAF, and log cluster size as predictors. The recombination coefficient is treated as an association rather than as evidence of linked selection.

S6.4 Consensus-versus-representative validation

For clusters represented by both an eMLG consensus and a representative marker, we compared the inferred sorting direction between representations. The two representations agreed on the parental direction for approximately 99% of these clusters, confirming that using representatives for genome-wide architecture models does not materially change the parental direction assigned to clusters, while the consensus remains preferable for aggregate sorting magnitude and cluster-level LD analyses.

S7 Association with climate

S7.1 Climate covariates

Nineteen bioclimatic variables were derived for each sampling location from monthly minimum temperature, maximum temperature, and precipitation data for 2020–2024 obtained from WorldClim v2.1 at 2.5-minute spatial resolution. Bioclimatic variables were calculated for each year using `bio_vars` in `dismo` v1.3.16 and averaged across the five years. The variables were summarised by principal-component analysis using `FactoMineR` v2.13. The first two axes had eigenvalues greater than one and together explained 86% of the climatic variation.

PC1 primarily described a latitudinal temperature and seasonality gradient, while placing the high-altitude Swiss population with colder northern populations. PC2 mainly described precipitation differences that separated the Swiss and Åland populations from the remaining populations. The PC1 and PC2 scores were used as population-level environmental covariates.

S7.2 BayPass configurations

Associations between population allele frequencies and PC1 or PC2 were evaluated using the standard covariate model in BayPass v3.1. Evidence was reported as Bayes factors on the deciban scale, BF(dB), with $\text{BF(dB)} \geq 15$ defining a significant marker-level association (Fig. S10).

The analysis comprised eight configurations crossing:

- the full population set and the set excluding Åland;
- PC1 and PC2;
- an among-population covariance matrix Ω estimated under the core model from the LD-pruned marker set and supplied to the covariate model (fixed- Ω), or Ω estimated internally from the full marker set.

The Åland-excluded, fixed- Ω configuration was designated primary. Full-set and internally estimated- Ω configurations were retained as sensitivity comparisons. No Sielva-excluded analysis was required for the current primary analysis. BayPass output rows were matched positionally to the input markers; each import asserted that row counts matched and indices were strictly increasing.

S7.3 Population structure and ancestry–climate association

For each hybrid population, mean *F. aquilonia* ancestry was estimated across parentally diagnostic markers. Its rank correlation with PC1 and PC2 was calculated across populations and reassessed after removing populations one at a time. In the current analysis, PC1 was weakly correlated with mean ancestry ($\rho = -0.23$), whereas PC2 showed a moderate positive correlation ($\rho = 0.57$) that was strongly influenced by Åland (Fig. S7).

This pattern motivated two controls: use of the covariance matrix Ω to account for shared population structure, and designation of the Åland-excluded analysis as primary. These controls reduce, but cannot demonstrate the complete absence of, confounding by population history. We therefore interpret genotype–environment associations as conditional associations rather than direct evidence of local adaptation.

S7.4 Cluster-level climate-association rate

For each LD cluster, climate-association strength was summarised as the proportion of member markers with $\text{BF(dB)} \geq 15$, hereafter the significance rate. A rate was used instead of a fixed number of significant members because a count imposes a much higher proportional requirement on small clusters than on large clusters. Cluster size was retained as a covariate for its independent association with the outcomes.

Because the significance rate is estimated with greater sampling error in small clusters, analyses were reported for nested strata comprising all clusters, clusters with at least 20 markers, and clusters with at least 50 markers. The strata were interpreted as precision analyses rather than as alternative significance thresholds.

S7.5 Diagnostic-index enrichment

For each cluster, the outcome was the proportion of member markers exceeding the ancestry-informative criterion $DI > -25$. Each cluster contributed one observation. This proportion was modelled against the climate-association rate, with cluster size included as a covariate (Fig. S8). A size-weighted model with an overdispersion correction was fitted as a complementary check. The cluster-level model is primary because a marker-level model would treat correlated members of the same cluster as independent observations.

S7.6 Sorting-climate overlap

The relationship between climate association and ancestry sorting was tested at the cluster level using the same significance-rate predictor and nested size strata. The outcome was the cluster-level sorting classification described above. The significance-rate analysis is primary; inference was not based on a fixed count of significant markers within a cluster.

The climate-association analysis therefore has the following inferential hierarchy:

1. $BF(dB) \geq 15$ defines marker-level climate association;
2. the cluster significance rate is the primary exposure;
3. the Åland-excluded, fixed- Ω configuration is primary;
4. the ≥ 20 - and ≥ 50 -marker strata evaluate precision;
5. alternative population sets and covariance treatments are sensitivity analyses.

S8 Dependence-aware uncertainty and climate sensitivity

S8.1 Genomic block bootstrap

Residual spatial dependence can remain after each LD cluster is represented once because neighbouring units may share ancestry tracts, recombination environment, and local diversity. We therefore recalculated uncertainty for five central coefficients using a non-parametric genomic block bootstrap: the association of DI with sorting direction, the associations of recombination with sorting direction and magnitude, and the associations of the PC1 and PC2 climate-significance rates with the proportion of ancestry-diagnostic markers.

The sorting-direction model was a logistic regression of *F. aquilonia*-directed sorting on standardised DI, standardised $\log_{10}(\text{recombination} + 0.1)$, standardised pooled parental MAF, and standardised log cluster size. Sorting magnitude was modelled against standardised $\log_{10}(\text{recombination} + 0.1)$ and standardised DI. These analyses used one representative per LD-reduced unit. The climate models used clusters with at least 50 markers from the primary Åland-excluded, fixed- Ω configuration and modelled the proportion of diagnostic members against the climate-significance rate and log cluster size.

Blocks were defined in two ways: whole chromosomes, giving 26 blocks in the analysed marker map, and contiguous 10-cM intervals within chromosomes. The latter approximately matched the inferred scale of admixture LD, whereas the chromosome analysis made no assumption about a shorter dependence scale. For each definition, blocks were sampled with replacement to their original number, all units in each sampled block were retained, and the relevant model was refitted. Central 95% percentile intervals were calculated from 10,000 replicates (Fig. S11).

Before resampling, every reconstructed model was required to reproduce its original full-data coefficient; failure of this check stopped the analysis.

S8.2 Allele-frequency and differentiation sensitivity of the climate analysis

BayPass evidence depends partly on the amount of allele-frequency variation among populations. A cluster with little variation has limited opportunity to show a genotype–environment association, whereas a differentiated cluster can carry both stronger association evidence and more ancestry-diagnostic markers. We therefore evaluated whether the nominal diagnostic-locus enrichment was sensitive to cluster-level summaries of allele-frequency variation and differentiation.

The analysis retained the primary Åland-excluded, fixed- Ω configuration and the 878 clusters containing at least 50 markers. For each climate axis, four nested linear models were fitted:

$$\begin{aligned} M0 : & f_{DI} \sim p_{\text{climate}} + \log(n_{\text{loci}}), \\ M1 : & M0 + \overline{\text{MAF}}_{\text{hybrid}}, \\ M2 : & M1 + \overline{\text{recombination}}, \\ M3 : & M2 + \overline{\text{XtX}}, \end{aligned}$$

where f_{DI} is the proportion of ancestry-diagnostic members and p_{climate} is the proportion of members with $\text{BF}(\text{dB}) \geq 15$. Hybrid MAF, recombination rate, and BayPass XtX were averaged across cluster members. The climate-rate coefficient was reported as the percentage-point change in diagnostic content per 10-percentage-point increase in climate-significance rate. The fully adjusted coefficient was additionally assigned a chromosome block-bootstrap 95% interval using 10,000 replicates (Fig. S9).

This model sequence was treated as a sensitivity analysis rather than a causal adjustment set. In particular, XtX and the covariate Bayes factor derive from the same population allele frequencies and are not independent; conditioning on XtX may remove biologically relevant differentiation as well as generic power-related variation. Attenuation across the sequence therefore tests whether the nominal enrichment can be distinguished from shared dependence on allele-frequency differentiation, rather than estimating a uniquely climate-mediated effect.

S9 Statistical interpretation

The analyses are descriptive and conditional on the sampled populations, quality gates, and analysis definitions. Consistent sorting can be produced by more than one evolutionary process, and a correlation between allele frequencies and climate is compatible with environmental selection but does not by itself establish it. We therefore use terms such as “associated”, “enriched”, and “consistent with”, reserving causal language for results that receive convergent support and biological scrutiny of candidates.

Block-robust intervals are treated as the principal uncertainty check for the central coefficients. The nested MAF, recombination, and XtX models are limited to diagnosing the nominal climate–diagnostic relationship and do not replace the simpler primary model or establish a causal partition of its coefficient.

S10 Software, reproducibility, and current status

LD decay estimation, clustering, and consensus construction were implemented using functions from `LDscnR`; project-level analyses were performed in R using scripts in the `formica_hybrid` repository. Analysis settings, thresholds, and intermediate checkpoints were retained with the corresponding outputs so that downstream analyses used the same marker partition.

Open item. Add the final `LDscnR` citation, version, and repository or archive identifier. Software versions not already given above should be added from the locked analysis environment before submission.

Table 1: Principal parameter settings used in the analyses.

Analysis	Parameter	Value
Marker set	Hybrid MAF threshold	≥ 0.05
	Retained markers	1,114,340
LD decay	Windows per chromosome	20, 50% overlap
	Marker pairs per window	$\leq 5,000$, 20 distance strata
	r^2 summary	0.95 quantile
	Pair MAF for curve fitting	> 0.1 for both members
Stage 1	Relative LD threshold	$\rho = 0.5$, per chromosome
	Clustering	connected components, then complete linkage
	Representative	highest median r^2 to cluster members
Stage 2	Flagging	$ld_w > 0.025$ or ≥ 5 markers
	Consensus fidelity	≥ 0.80
	Pairwise correlation floor	$r^2 \geq 0.2$
	Distance restriction	0.5 cM
	Stored consensus	clusters with ≥ 5 markers
Sorting	Near-fixation tolerance	$\theta = 0.15$
	Classification threshold	$\tau = 0.5$
	Parental MAF gate	≥ 0.15
	Minimum near-fixed populations for test	5
	Directional null	$p = 0.5$
	Multiple-testing adjustment	Benjamini–Hochberg
Climate	Marker evidence threshold	$BF(dB) \geq 15$
	Primary population set	Åland excluded
	Primary covariance treatment	fixed Ω , estimated from LD-pruned markers
	Cluster exposure	significance rate
	Precision strata	all, ≥ 20 , and ≥ 50 markers

Analysis	Parameter	Value
	DI criterion	DI > -25
Block bootstrap	Replicates	10,000
	Block definitions	chromosome (26 blocks) and contiguous 10-cM intervals
	Interval	central 95% percentile
Climate sensitivity	Analysis stratum	primary configuration, clusters with ≥ 50 markers
	Covariate sequence	hybrid MAF, recombination rate, and BayPass XtX
	Fully adjusted uncertainty	chromosome block bootstrap, 10,000 replicates

References

- [1] Nouhaud P, Martin SH, Portinha B, Sousa VC, Kulmuni J. Rapid and predictable genome evolution across three hybrid ant populations. *PLoS Biology*. 2022;20:e3001914. doi:10.1371/journal.pbio.3001914.